

Make notes Very detailed (full sentences, no short phrases)

- ✓ Including all important theoretical points
- ✓ Including procedural steps wherever steps are given
- ✓ Formatted cleanly for exam

make sure to include each and every imp point.



Below are **very detailed, cleanly formatted exam-ready notes** for **Module 3: AWS Global Infrastructure Overview**, covering **every important theoretical point and procedural step** from the provided material.

(Content prepared directly from the uploaded module  AcademyCloudFoundations_Module_...)

MODULE 3 – AWS GLOBAL INFRASTRUCTURE OVERVIEW (DETAILED NOTES)

1. AWS GLOBAL INFRASTRUCTURE – INTRODUCTION

The AWS Global Infrastructure is designed and built to deliver a **flexible, reliable, scalable, and secure cloud computing environment** with **high-quality global network performance**.

AWS continuously expands and updates its infrastructure footprint across the world to meet customer demand.

The global infrastructure consists mainly of:

- AWS Regions
- Availability Zones
- Edge Locations
- Regional Edge Caches
- Local Zones

These components together ensure **low latency, fault tolerance, scalability, and high availability**.

2. AWS REGIONS

An **AWS Region** is a **geographical area** where AWS has multiple data centers.

Key characteristics of AWS Regions:

- Each Region is completely independent and isolated from other Regions.
- Data replication across Regions is fully controlled by the customer.
- Communication between Regions takes place over AWS's high-speed private backbone network.
- Each Region provides full redundancy and strong network connectivity.
- A Region typically consists of **two or more Availability Zones**.
- AWS operates Regions across multiple continents worldwide.

Factors for selecting a Region:

1. **Data governance and legal requirements** – Some countries require data to stay within geographic boundaries.
 2. **Proximity to customers** – Choosing a closer Region reduces latency.
 3. **Services availability** – Not all AWS services are available in every Region.
 4. **Cost differences** – Pricing may vary slightly between Regions.
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3. AVAILABILITY ZONES (AZs)

An **Availability Zone** is a **fully isolated portion of a Region**.

Important features:

- Each Region contains multiple Availability Zones.
- Each Availability Zone consists of **one or more discrete data centers**.
- AZs are designed specifically for **fault isolation**.
- They are interconnected with **high-speed private fiber networks**.
- Customers choose which AZs their resources run in.
- AWS strongly recommends distributing resources across multiple AZs for resiliency.

Purpose:

Availability Zones help in building applications that are:

- ✓ Highly available
 - ✓ Fault tolerant
 - ✓ Protected from single point failures
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4. AWS DATA CENTERS

AWS data centers are the **physical buildings** where computing actually happens.

Characteristics:

- Designed with very high security standards.
 - Store customer data and run processing workloads.
 - Each data center has:
 - Redundant power supply
 - Backup generators
 - Redundant networking
 - Strong cooling systems
 - Each data center is housed in a separate physical facility.
 - Typically contains **50,000 to 80,000 physical servers**.
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5. POINTS OF PRESENCE (PoP)

AWS maintains a global network of **Points of Presence**.

These include:

➤ Edge Locations

➤ Regional Edge Caches

They are mainly used by **Amazon CloudFront (AWS Content Delivery Network)**.

Purpose:

- Deliver content such as videos, images, websites, and APIs with **very low latency**
 - Cache content closer to users
 - Reduce load on origin servers
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6. REGIONAL EDGE CACHES vs EDGE LOCATIONS

Edge Locations:

- Closest locations to end users
- Store frequently accessed content
- Requests automatically routed to nearest edge location
- Not tied to Regions or AZs

Regional Edge Caches:

- Larger storage capacity than individual edge locations
 - Store content for longer durations
 - Positioned between origin servers and edge locations
 - Reduce repeated fetching from original servers
 - Improve overall CloudFront performance
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7. LOCAL ZONES

Local Zones extend AWS services **closer to large metropolitan areas**.

Key points:

- Provide ultra-low latency for latency-sensitive applications
- Offer select AWS services such as:
 - EC2
 - EBS
 - VPC
 - Used for workloads like gaming, real-time streaming, and high-performance computing
 - Connected to the parent AWS Region using private AWS network
 - Other services such as S3 and Aurora are accessed privately through VPC

Comparison:

Feature	Local Zones	Edge Locations
Purpose	Run workloads	Cache content
Services	Compute & storage	Content delivery
Deployment	Metro areas	Worldwide
Main use	Low latency apps	Faster downloads



8. AWS INFRASTRUCTURE DESIGN FEATURES



Elasticity:

The infrastructure automatically adjusts resources up or down based on demand.



Scalability:

The system can grow to accommodate increasing workloads without performance loss.



Fault Tolerance:

AWS continues operating even when components fail through built-in redundancy.



High Availability:

Ensures minimal downtime by distributing workloads across AZs and Regions.



9. AWS SERVICE CATEGORIES (OVERVIEW)

AWS services are grouped into organized categories such as:

- Compute
- Storage
- Databases
- Networking and Content Delivery
- Security, Identity, and Compliance
- Management and Governance

- Cost Management
 - Analytics
 - Machine Learning
 - Application Integration
 - Mobile Services
 - Migration and Transfer
 - Business Applications
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10. STORAGE SERVICES

Main storage services:

- Amazon S3 – Object storage for scalable data storage
 - Amazon EBS – Block storage for EC2 instances
 - Amazon EFS – File storage shared across EC2 instances
 - Amazon S3 Glacier – Low-cost archival storage for long-term backups
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11. COMPUTE SERVICES

- Amazon EC2 – Virtual servers
 - EC2 Auto Scaling – Automatically adjusts server count
 - AWS Lambda – Serverless computing
 - Elastic Beanstalk – App deployment platform
 - Amazon ECS – Container management
 - Amazon EKS – Kubernetes service
 - AWS Fargate – Serverless container engine
 - Amazon ECR – Container registry
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12. DATABASE SERVICES

- Amazon RDS – Managed relational databases
- Amazon Aurora – High-performance relational database
- Amazon DynamoDB – NoSQL key-value database
- Amazon Redshift – Data warehouse service



13. NETWORKING & CONTENT DELIVERY

- Amazon VPC – Private cloud network
 - Elastic Load Balancing – Distributes traffic
 - Amazon CloudFront – Content delivery network
 - Amazon Route 53 – DNS service
 - AWS Direct Connect – Dedicated network link
 - AWS VPN – Secure remote access
 - AWS Transit Gateway – Connects multiple networks
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14. SECURITY, IDENTITY & COMPLIANCE

- AWS IAM – User access control
 - AWS Organizations – Manage multiple accounts
 - Amazon Cognito – User authentication
 - AWS Artifact – Compliance reports
 - AWS KMS – Encryption key management
 - AWS Shield – DDoS protection
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15. COST MANAGEMENT

- AWS Cost Explorer – Analyze spending
 - AWS Budgets – Set spending alerts
 - AWS Cost & Usage Report – Detailed billing data
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16. MANAGEMENT & GOVERNANCE

- AWS Management Console – Web interface
- AWS CLI – Command line control
- AWS CloudWatch – Monitoring and logs
- AWS CloudTrail – Track API activity
- AWS Config – Resource tracking
- AWS Trusted Advisor – Best practice checks

- AWS Well-Architected Tool – Architecture review
 - AWS Auto Scaling – Automatic scaling of services
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17. HANDS-ON ACTIVITY – AWS CONSOLE (PROCEDURAL STEPS)

Step 1:

Launch AWS Management Console in the sandbox environment.

Step 2:

Click **Services Menu** and observe service categories.

Step 3:

Locate:

- IAM under **Security, Identity, and Compliance**
- Amazon VPC under **Networking & Content Delivery**

Step 4:

Open Amazon VPC and observe Region selector at top right.

Step 5:

Change Region (for example from N. Virginia to London).

Step 6:

Click **Subnets** and select one subnet.

👉 Observation: Subnets exist at **Availability Zone level**.

Step 7:

Click **Your VPCs**

👉 Observation: VPCs exist at **Region level**.

Step 8:

Identify global services:

- ✓ IAM and Route 53 are global
 - ✓ EC2 and Lambda are Regional
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18. KEY TAKEAWAYS (VERY IMPORTANT FOR EXAM)

- AWS Global Infrastructure consists of Regions and Availability Zones.
 - Regions are chosen based on compliance, latency, service availability, and cost.
 - Availability Zones are isolated and fault-tolerant.
 - Edge locations and Regional Edge Caches improve performance by caching content close to users.
 - Local Zones provide ultra-low latency computing near metropolitan areas.
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If you'd like, I can now:

- ✓ Convert these into **short revision sheets**
- ✓ Add **memory tricks & mnemonics**
- ✓ Prepare **5-mark and 10-mark exam answers**
- ✓ Create **last-day quick review notes**

Just tell me 👍

Sources

